How-To Library

Informatica

Performing Domain Discovery on a Microsoft SharePoint Source Using Informatica Discovery Agent in Data Privacy Management

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Abstract

This article describes the steps to perform domain discovery on a Microsoft SharePoint source using Informatica Discovery Agent in Data Privacy Management. The article includes all steps required to create an Informatica Discovery Agent, perform a scan, and view the results.

Supported Versions

- Data Privacy Management 10.4.1

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Overview

You can use an Informatica Discovery Agent to perform a domain discovery scan on semi-structured and unstructured sources in Data Privacy Management.

You can configure a scan to read all folders, read specific folders from a list, or read folders based on regular expressions.

To use an Informatica Discovery Agent to perform a scan on a Microsoft SharePoint source, perform the following high-level steps:

1. Install and configure the Informatica Discovery Agent for domain discovery and start the agent.
2. Create a connection to the agent.
3. Create a classification policy.
4. Create a Microsoft SharePoint data store.
5. Update the Informatica Discovery Agent.
6. Create and run the scan.
7. Monitor the scan job.
8. View the scan results.

Note: You can install and run the Informatica Discovery Agent only on a Linux machine.

Step 1. Install Informatica Discovery Agent

The Informatica Discovery Agent installer is included in the Data Privacy Management files.

Complete the following steps to install the Informatica discovery agent:

1. Perform the prerequisite tasks.
2. Install the Informatica Discovery Agent.
3. Configure the `server.xml` file to configure the agent.

Prerequisites

Complete the following tasks before you install the agent:

1. Generate a keystore file for the Informatica Discovery Agent.
2. Import the Informatica Discovery Agent keystore certificate into the Informatica domain truststore.
3. Import the Data Privacy Management Service certificate into the Informatica Discovery Agent truststore. The following example explains the steps to generate keystore and trustore files for the agent:
    1. Create a `DiscoveryAgentcerts` folder.

        ```
        mkdir DiscoveryAgentcerts
        ```
    2. Browse to the folder: `cd DiscoveryAgentcerts`
    3. To generate the keystore file, run the following command:

        ```
        keytool -genkey -alias <Certificate_Alias> -keystore <agent_keystore_file_name>.jks -keyalg RSA -sigalg SHA1withRSA -storepass <agent keystore password> -dname CN=<fully qualified hostname>,OU=Informatica,O=Informatica,L=RedwoodCity,S=California,C=US -validity <number_of_days_the_key_is_valid>
        ```

        Replace placeholders in the command with required values.
    4. Press **Enter** at the password prompt.
    5. To export the certificate, run the following command:

        ```
        keytool -export -alias <Certificate_Alias> -file <agent_certificate_filename>.cer -keystore <agent_keystore_file_name>.jks
        ```
    6. Press **Enter** at the password prompt.
    7. If the Informatica Discovery Agent is installed on a different machine, copy `<Informatica installation directory>/tomcat/conf/infa_dflt.cert` from the Data Privacy Management domain machine to the Informatica Discovery Agent machine.
    8. To import the Data Privacy Management Service into the Informatica Discovery Agent truststore, run the following command:

        ```
        keytool -import -alias <DPM_certificate_Alias> -file <Data Privacy Management Service certificate path>/infa_dflt.cert -keystore <agent_trustore_file_name>.jks -storepass <agent truststore password>
        ```
    9. At the prompt to trust the certificate, enter **Yes**.
    10. If the Informatica Discovery Agent is installed on a different machine, copy `<Informatica installation directory>/tomcat/conf/infa_dflt.cert` from the Data Privacy Management domain machine to the Informatica Discovery Agent machine
    11. To import the Informatica Discovery Agent certificate into Data Privacy Management Domain truststore, run the following command:

```
<Informatica installation directory>/java/jre/bin/keytool -import -alias
<agent_certificate_Alias> -file <agent_certificate_filename>.cer -keystore <Informatica
installation directory>/services/shared/security/infa_truststore.jks -storepass <dpm
truststore password>
```

12. At the prompt to trust the certificate, enter **Yes**.
    The certificates are generated and imported successfully.

Extract the Agent Files to Install the Agent

The Informatica Discovery Agent installer is included in the Data Privacy Management files.

1. Navigate to the following folder: `<Informatica installation directory>/SecureAtSourceService/InformaticaDiscoveryAgent`

   ```
   cd <Informatica installation directory>/SecureAtSourceService/InformaticaDiscoveryAgent
   ```

2. Extract the contents of the `InformaticaDiscoveryAgent.zip` file to a folder in any location on the machine. For example: `Discovery Agent`

   ```
   unzip InformaticaDiscoveryAgent.zip -d /<Discovery Agent>
   ```

Update the Server Configuration File

Update the server port and truststore and keystore details in the following server configuration file: `server.xml`

1. Navigate to the following folder: `<Discovery Agent>/tomcat/conf`
2. Open the `server.xml` file.
3. Locate the following lines in the text:

   ```
   <Connector port="8080" protocol="HTTP/1.1"
   connectionTimeout="20000"
   redirectPort="8443">
   ```

4. Comment out the lines.

   Add `<!--` before the section and `-->` after the section.

   ```
   <!-- <Connector port="8080" protocol="HTTP/1.1"
   connectionTimeout="20000"
   redirectPort="8443" /> -->
   ```

5. Find a port that is available for the agent.

   You can use the netstat command to determine port availability. For example, use the following command to determine if port 9000 is available:

   ```
   netstat -anp | grep 9000
   ```

6. Add the following configuration information:

   ```
   <Connector port="<port number>" protocol="org.apache.coyote.http11.Http11NioProtocol"
   maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
   truststoreFile="<discovery agent truststore file path>" truststorePass="<truststore
   password>"
   keystoreFile="<discovery agent keystore file path>" keystorePass="<keystore password>"
   keyAlias="<alias name of the certificate used for the discovery agent>"
   clientAuth="true" sslProtocol="TLS" />
   ```

7. Save the changes to `server.xml` file.

Start or Stop the Informatica Discovery Agent

You can start up and shut down the Informatica Discovery Agent. Before you run a scan, the agent must be up and running.

1. Browse to the `bin` folder in the Informatica Discovery Agent folder.
   `cd <Informatica Discovery Agent folder>/bin`
2. Run the required command:
   - Start the agent: `./siagent.sh startup`
   - Shut down the agent: `./siagent.sh shutdown`

Step 2. Create a Connection to the Informatica Discovery Agent

After you install and configure the Informatica Discovery Agent, you must create a connection to the agent in Data Privacy Management.

1. Log in to Data Privacy Management.
2. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace appears.
3. To create a remote agent, select **New** > **Subject Registry Agent** from the **Actions** menu.

   The **New_Subject_Registry_Agent** page appears.
4. On the **Specify Properties** pane, enter a name for the remote agent.

   The name cannot exceed 255 characters.
5. Optionally, enter a description.

   The description cannot exceed 255 characters.
6. Enter the host address of the remote agent server.
7. Enter the admin port number of the remote agent at the specified host address.
8. Optionally, test the connection to the remote agent server.
9. Click **Save**.

Step 3. Create a Classification Policy

A classification policy defines the rules that identify sensitive data and classify a data store based on industry or organization data security standards. You use a classification policy in a scan. You can use a system-defined classification policy or create a classification policy.

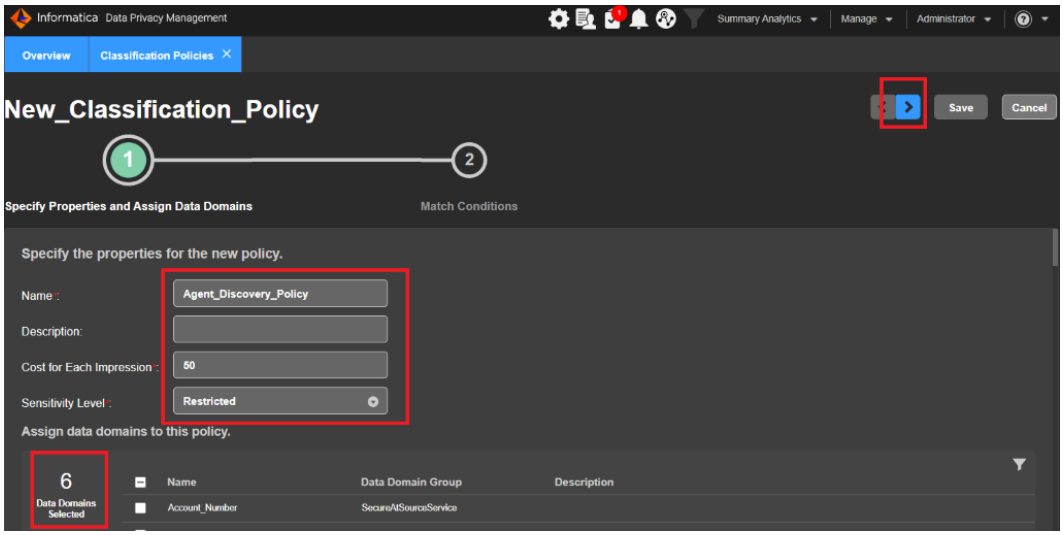
You can use Enterprise Keywords from Microsoft SharePoint as file tags. You can use Enterprise Keywords from Microsoft SharePoint lists. You cannot use Enterprise Keywords from Microsoft SharePoint libraries. Verify that the Enterprise Keywords that you use as file tags are created appropriately in Microsoft SharePoint.

1. Click **Manage** > **Classification Policies**.

   The **Classification Policies** workspace appears.
2. Click **New**.

   The **New_Classification_Policy** page appears.
3. On the **Specify Properties and Assign Data Domains** panel, enter values for the properties.

The following image shows the **New Classification Policy** page with the properties entered:



4. Select the data domains that you want to include in the classification policy.

   In this example, we select the following data domains:

   - City
   - CreditCardNumber
   - FirstName
   - LastName
   - SSN_General
   - USZip_5digit

5. Click the **Next** icon.

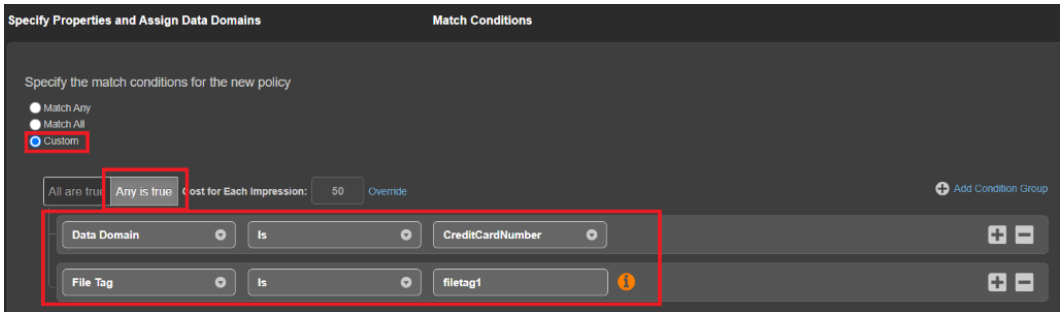
   The **Match Conditions** page appears.

6. Specify a match condition.

   Add the following Custom match condition:

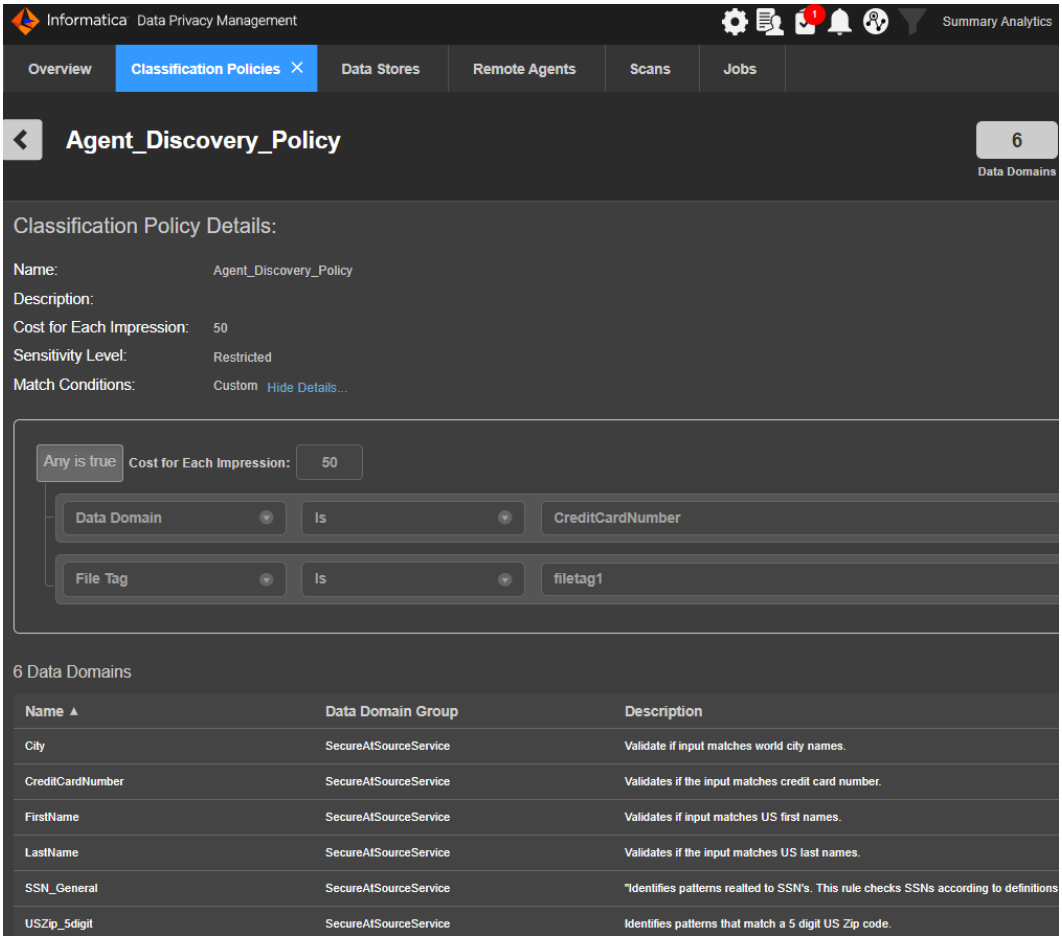
   ```
   (Data Domain is  CreditCardNumber)
   OR
   (File Tag is filetag1)
   ```

   The following image shows the **Match Conditions** Page:



7. Click **Save**.

   The following image shows the **Details** page of the Agent_Discovery_Policy that we created:



Step 4. Create a Microsoft SharePoint Data Store

A data store defines a connection to a data source. Create a data store for each data source that contains sensitive data that you want to discover.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New_Data_Store** page appears.

3. Enter the connection properties for the data store.

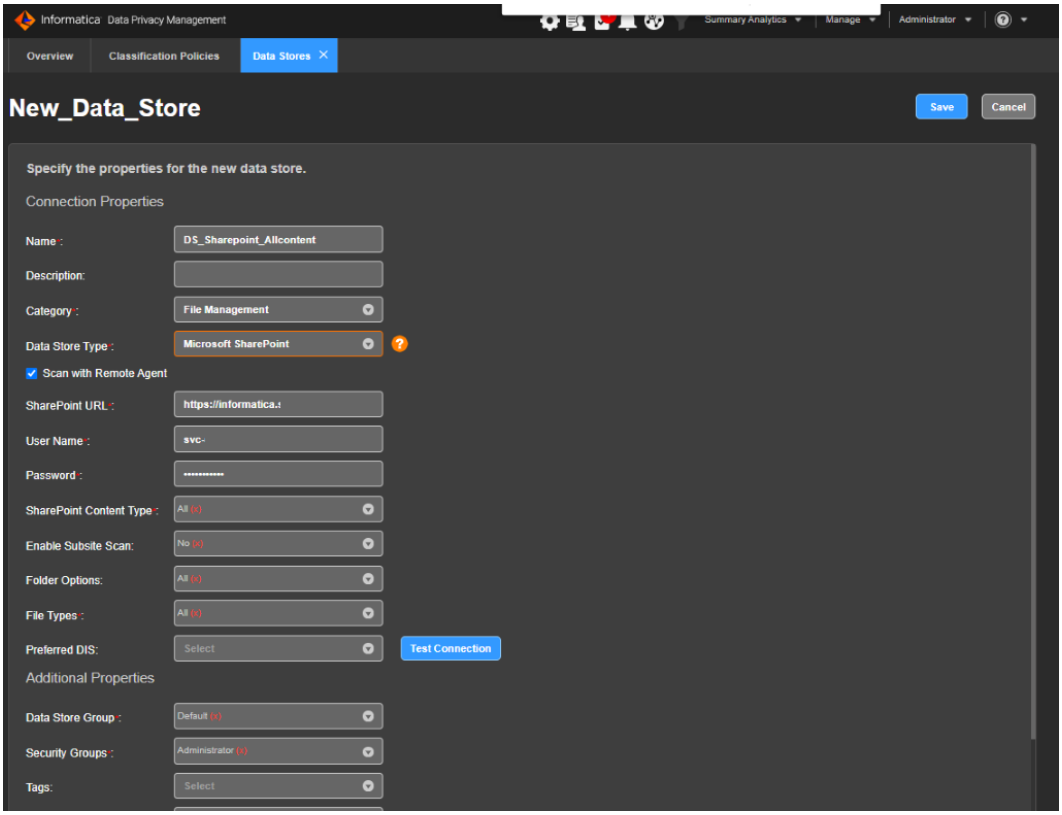
   Choose the following options:

   - Category: File Management
   - Data Store Type: Microsoft SharePoint
   - Scan with Remote Agent

4. Enter the SharePoint URL, and the user name and password.

5. Retain default settings for all other properties.

   The following image shows the **New Data Store** page with all details entered:



Step 5. Update the Informatica Discovery Agent

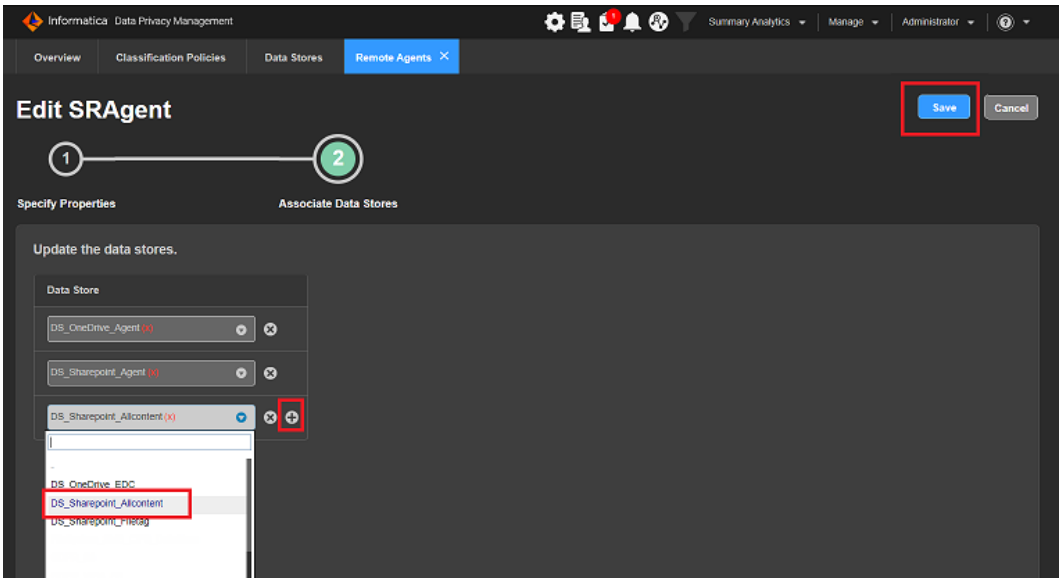
After you create the data store, add it to the list of associated data stores in the Informatica Discovery Agent.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.
2. Select the checkbox next to the remote agent that you want to edit.
3. From the **Actions** menu, select **Edit**.

   The **Edit_<Remote Agent Name>** page appears.
4. Open the **Associate Data Stores** section.
5. Associate the data store with the remote agent. Click the **Add** button and select the Microsoft SharePoint data store from the list.

The following image shows the list of data stores on the **Associate Data Stores** page:



6. Click **Save**.

Step 6. Create and Run the Scan

Create a scan to configure how and when the scan job discovers and classifies sensitive data. When you create a scan, you specify the type of scan, configure the options, and schedule when the scan runs.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.

2. Click **New**.

   The **Define & Connect** panel appears.

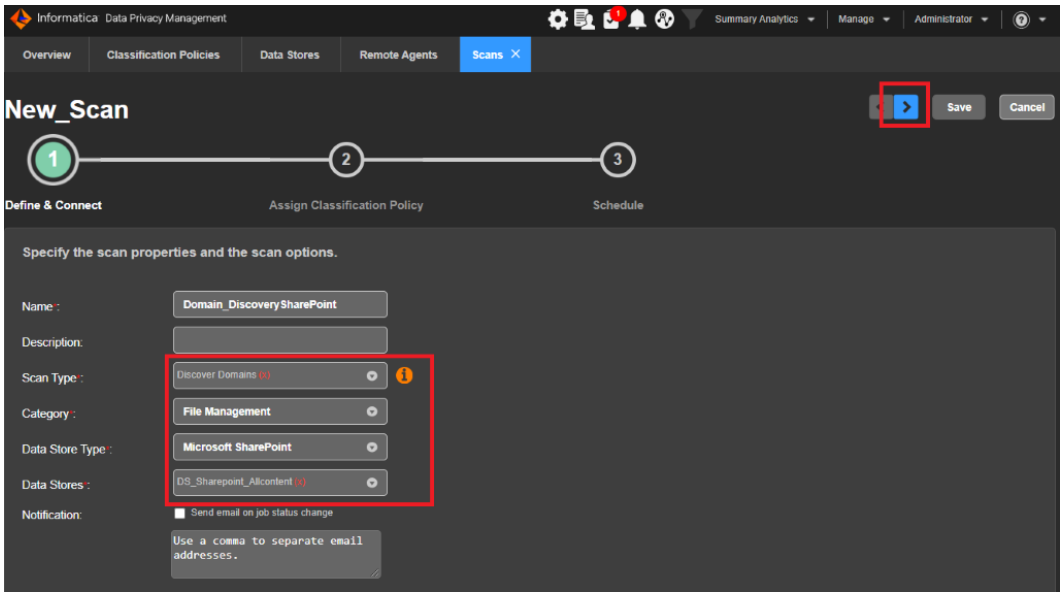
3. Enter a name and optional description of the scan.

4. Choose the **Discover Domains** scan type.

   A Discover Domains scan maps fields in data stores to data domains that match the classification policies you include in the scan.

5. Choose the File Management category and the Microsoft SharePoint data store type, and choose the data store from the list of data stores.

The following image shows the **New Scan** page with information entered:

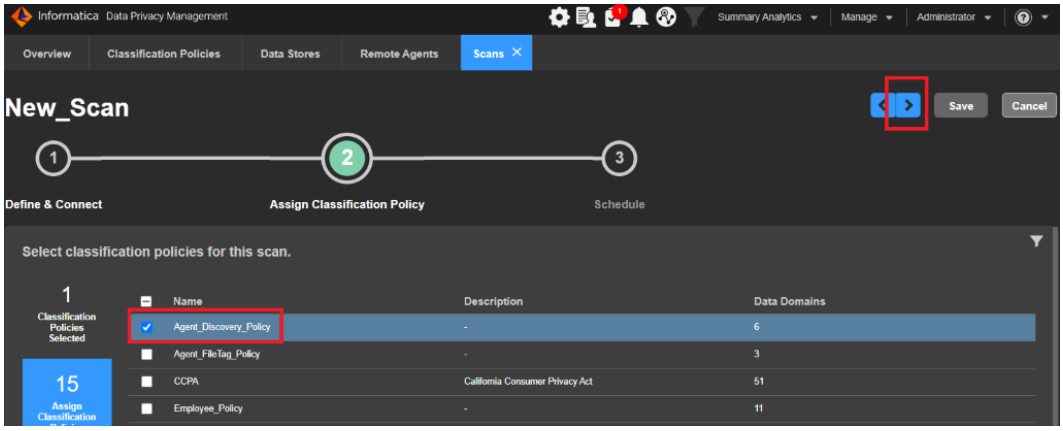


6. Click **Next**.

   The **Assign Classification Policy** panel appears.

7. Select the classification policy created.

   The following image shows the **Assign Classification Policy** panel with the policy selected:



8. Click **Next**.

   The **Schedule** panel appears.

9. Select **Run Now** to run the scan immediately.

10. Click **Save**.

    The scan job is submitted.

You can monitor the job to verify that the scan completes successfully.

Step 7. Monitor the Scan Job

Monitor the scan job to verify that the job completes successfully.

1. Click **Manage** > **Jobs**.

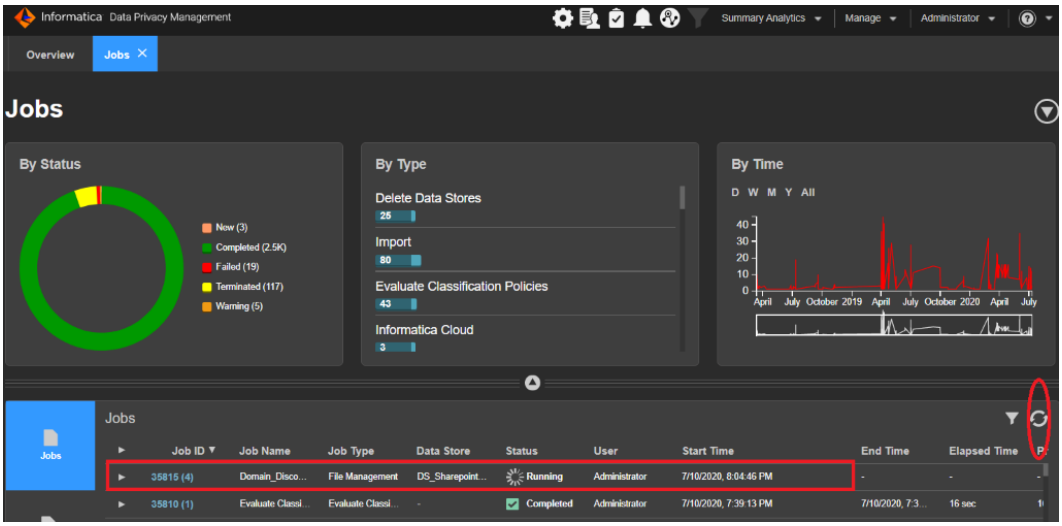
   The **Jobs** workspace appears.

2. Select the job.

   To select a job, click any job property except for the Job ID.

3. Verify the status of the job. Click **Refresh** or enable **Auto Refresh** to view the latest status.

   The following image shows the **Jobs** workspace with the job in Running state:



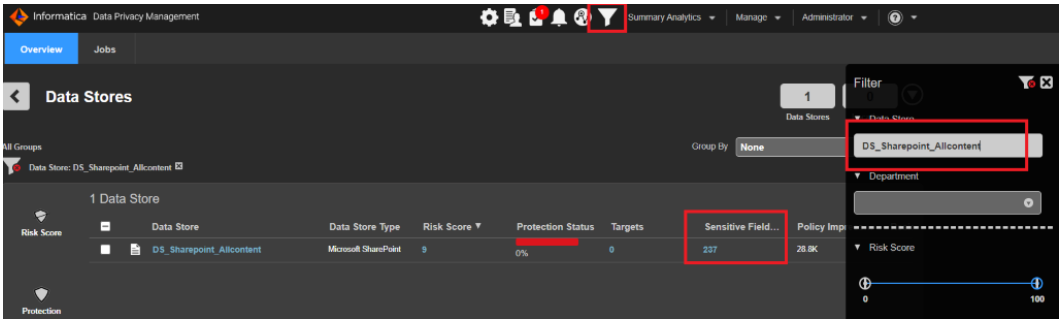
When the job completes successfully, you can review the scan results.

Step 8. View Scan Results

You can view the results of a successful scan job.

1. On the **Overview** section of the **Security Dashboard**, open the **Top Data Stores** widget.
2. Click the global filter and choose the SharePoint data store from the list of data stores.

   The following image shows the global filter icon on the **Data Stores** page:



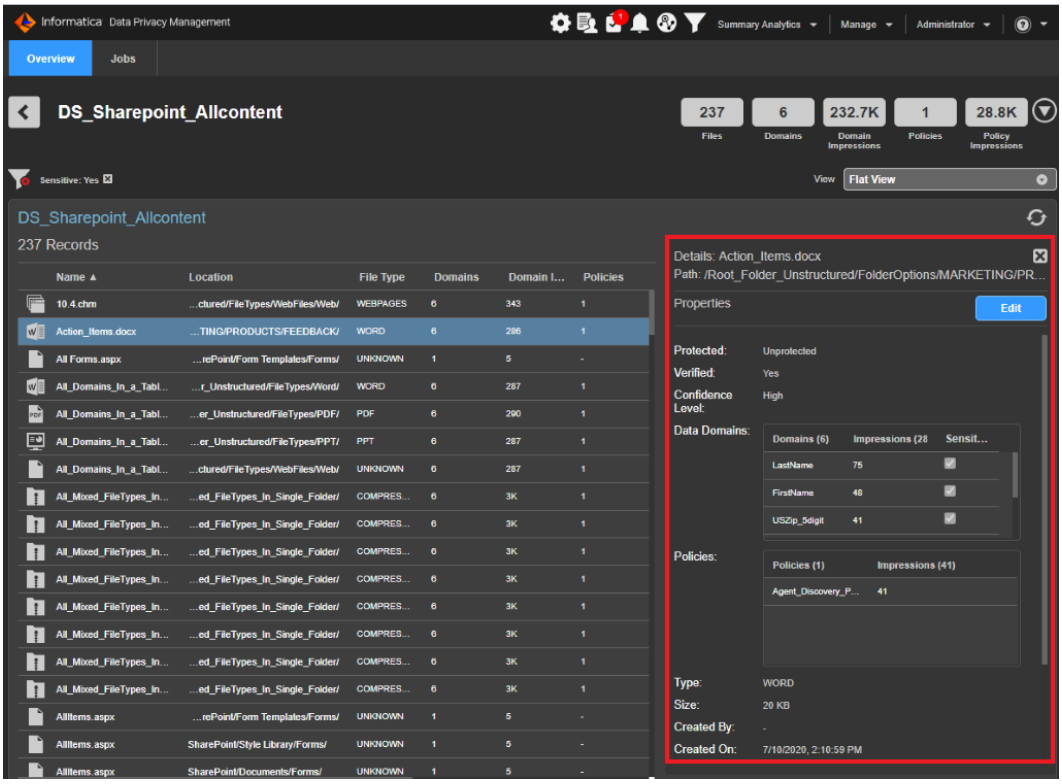
   The data store entry appears in the list of data stores.

3. Click the entry in the **Sensitive Fields** column.

   The **Sensitive Fields** page opens on the default Flat View. You can verify the file sensitivity, Impression, Confidence Level, and Verified Flag information.

4. Click a file name from the list of files to open the **Details** panel.

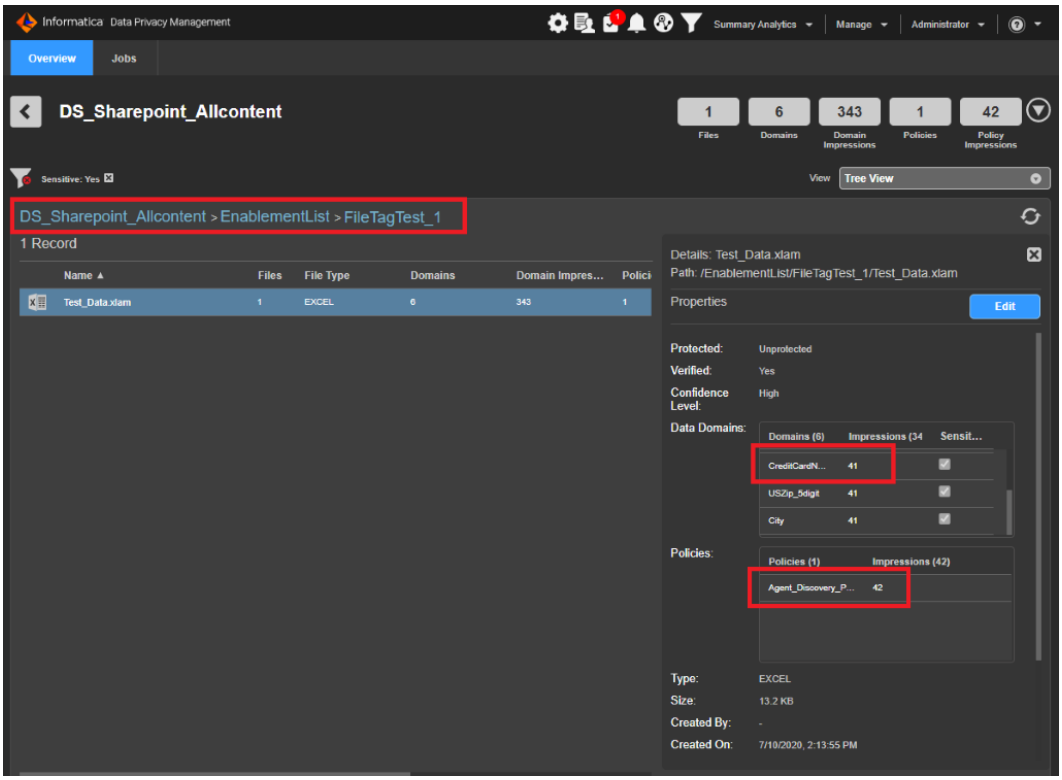
   The following image shows the Flat View with the **Details** panel visible:



   You can verify the accuracy of a domain for a sensitive field and verify the Domain Impression information.

5. To change the view to Tree View, select Tree View from the **View** list.
6. Verify the tag functionality. Navigate until the file level on a folder.

The following image shows the Tree View open at the file level, with the **Details** panel open:



Review the Policy Impression information. The Policy Condition is `(Domain is CreditCardNumber) OR (FileTag is filetag1)`. If the file contains the specified file tag, the policy impression increases by 1.
`Policy impression = Domain impression of CreditCardNumber Domain + 1 = 41 +1 = 42`

Note:

- If the file matches the classification policy condition that includes file tags, the policy impression for the file increases by 1.
- If you use a policy that includes file tags with Microsoft SharePoint and use Enterprise Data Catalog to run the scan, the policy impression might not appear as expected. If you use Enterprise Data Catalog to run the scan, use a classification policy without file tags.

Author

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